

Elevated Temperature Affects Development, Survivorship, and Settlement of the Elkhorn Coral, *Acropora palmata* (Lamarck 1816)

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Abstract. Elevated seawater temperatures during the late summer have the potential to negatively affect the development and survivorship of the larvae of reef corals that are reproductive during that time of year. *Acropora palmata*, a major Caribbean hermatype, reproduces annually during August and September. *A. palmata* populations have severely declined over the past three decades, and recovery will require high recruitment rates. Such recruitment will be limited if larval supply is reduced by elevated temperatures. The effects of elevated temperatures on development, survival, and larval settlement of *A. palmata* were investigated by culturing newly fertilized eggs at temperatures ranging from 27.5 to 31.5 °C. Development was accelerated and the percentage of developmental abnormalities increased at higher temperatures. Embryo mortality peaked during gastrulation, indicating that this complex developmental process is particularly sensitive to elevated temperatures. Larvae cultured at 30 and 31.5 °C experienced as much as an 8-fold decrease in survivorship compared to those at 28 °C. Additionally, settlement was 62% at 28 °C compared to 37% at 31.5 °C. These results indicate that embryos and larvae of *A. palmata* will be negatively affected as sea surface temperatures continue to warm, likely reducing recruitment and the recovery potential of *A. palmata* on Caribbean reefs.

Introduction

The elkhorn coral, *Acropora palmata* (Lamarck), is an important reef building coral in the Caribbean. This fast

growing, branching coral dominated the shallower reef zones and framework construction of Caribbean reefs for much of the Pleistocene (Shinn, 1976; Lighty *et al.*, 1982) and its origin has been traced back to as early as the late Pliocene (McNeill *et al.*, 1997; van Oppen *et al.*, 2001). Throughout its range, over the last several decades *A. palmata* has experienced die-offs and decreases in population size caused by unprecedented epidemics of diseases such as white band and white pox (Aronson and Precht, 2001; Patterson *et al.*, 2002), outbreaks of corallivorous snails (Bruckner, 2000; Miller *et al.*, 2002; Baums *et al.*, 2003), and occasional bleaching (Lundgren and Hillis-Starr, 2008). Such losses, in addition to natural damage from hurricanes and tropical storms, has led to a decrease of about 80% in live *A. palmata* cover on many reefs (Gardner *et al.*, 2003; Glynn, 2004; Baums *et al.*, 2005). Concern about the major regional decline in the abundance of this critical hermatypic coral species has resulted in its designation as a “threatened” species under the U.S. Endangered Species Act (Department of Commerce NOAA, 2006). The return of large, healthy populations of *A. palmata* is vital to the future health and growth of Caribbean coral reefs as they are challenged by elevated temperatures, increased frequency and severity of major storms, rising sea level, and ocean acidification—the predicted consequences of climate change (Hoegh-Guldberg *et al.*, 2007; Bates *et al.*, 2008). High levels of sexual recruitment will be necessary for population recovery, and sexual recruitment, in turn, depends on larval supply (Hughes *et al.*, 1999, 2000).

Until recently, recruitment dynamics and larval biology of *A. palmata* have been largely understudied. Fragmentation, a form of asexual reproduction, has been a common way for *A. palmata* to maintain large, dense populations, and to some extent this led to the perception that sexual

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Abbreviations: CCA, crustose coralline algae.

reproduction was less important to the population dynamics of this species (Bak and Engel, 1979; Highsmith, 1982; Lirman, 2000). Recent studies, however, have brought attention to the important role that sexual recruitment plays for *A. palmata* (Miller, 2002; Baums *et al.*, 2005; Zubillaga *et al.*, 2008; Williams *et al.*, 2008).

Sexual recruitment for *A. palmata* begins with the broadcast spawning of gamete bundles over a few nights after the full moons of August and/or September (Szmant, 1986; Miller and Szmant, 2006), and subsequent fertilization and development in the plankton of planula larvae. *A. palmata* larvae are not competent to settle until after at least 4 to 8 days of development (Szmant and Miller, 2005; Miller and Szmant, 2006; Zubillaga and Szmant, unpubl. data), during which they are exposed to predation and environmental stressors, as well as to dispersal by currents. Factors that reduce survivorship during the planktonic phase will reduce larval supply. Larval supply is a major determinant of recruit density on the Great Barrier Reef (Hughes *et al.*, 1999, 2000) and would be expected to be similarly important on Caribbean reefs. Clearly, a better understanding of all aspects of the reproductive biology of *A. palmata* is necessary to evaluate the potential for maintenance and recovery of *A. palmata* populations on Caribbean reefs.

A serious consequence of climate change and global warming could be the effect of elevated seawater temperatures on the development and survival of the larval stages of tropical marine organisms already living near their thermal maxima (Hoegh-Guldberg, 1999; Munday *et al.*, 2009). During exceptionally warm years in recent decades, temperatures during the late summer have been one or more degrees Celsius above normal (Glynn and D'Croz, 1990; Glynn, 1991, 1996; Brown, 1997). For adult reef corals, prolonged elevated summer temperatures have been pointed to as the main cause of the phenomenon known as coral bleaching: high temperatures in combination with high light stress results in the loss of the corals' algal symbionts (Jokiel and Guinther, 1978; Glynn, 1996, 2004; Brown, 1997; Hoegh-Guldberg, 1999). Many corals die or suffer partial mortality during bleaching events, and bleaching is often followed by a higher incidence of disease (Harvell *et al.*, 2002). Adult colonies of *A. palmata* have experienced bleaching events (Lundgren and Hillis-Starr, 2008), but thus far have been less susceptible to temperature stress than other Caribbean corals such as *Montastraea* spp. and *Millepora* spp. (Wood *et al.*, 2008).

The current emphasis on understanding coral bleaching attributes the disorder to the effects of elevated temperature on the production of harmful superoxide radicals by the algal symbionts (Lesser, 1996, 1997), but high temperature is also harmful to the coral host independent of the effect on the algal symbionts (Dunn *et al.*, 2002, 2004; Baird, *et al.*, 2006). Many important Caribbean reef building corals reproduce during the warmest summer months, releasing

spawn into the warm water column (Szmant, 1986; Szmant and Miller, 2005). These are life stages that do not have algal symbionts but have been shown to be sensitive to elevated temperature. To date, there are no studies of how elevated temperatures affect the development of *A. palmata*.

The life stages of reproduction and development are often less tolerant than adults of temperature stress (Cossins and Bowler, 1987; Johnston and Bennett, 1996). Elevated seawater temperatures have been reported to negatively impact embryogenesis, larval development, and larval survival, and to reduce larval settlement of various scleractinian corals (Edmunds *et al.*, 2001, 2005; Bassim *et al.*, 2002; Nozawa and Harrison, 2002, 2007; Bassim and Sammarco, 2003; Edmunds, 2005; Baird *et al.*, 2006; Negri *et al.*, 2007; Randall and Szmant, 2009). Increased mortality, premature metamorphosis, and depressed photosynthesis have been observed as effects of elevated seawater temperatures on planulae of *Porites astreoides* (Edmunds *et al.*, 2001). Developmental aberrations (Bassim *et al.*, 2002), slowed or arrested development, and altered ciliary movement, phototactic responses, swimming/searching behavior, and reduced settlement (Bassim and Sammarco, 2003) have all been observed in larvae of *Diploria strigosa* in response to elevated temperatures.

Thus, it is clear that larvae of several scleractinian corals are negatively affected by the levels of elevated summer seawater temperatures (30–32 °C) observed during recent years with record high global temperatures. It is also evident that larvae of different species have a range of responses and sensitivities to elevated temperatures. Negri *et al.* (2007) found that some Pacific species of *Acropora* are more sensitive to elevated temperatures than other genera examined: *A. millepora* larvae displayed significantly reduced fertilization success and a higher frequency of embryonic abnormalities at 32 °C, but larvae of two other genera showed no response. What remains unanswered is how *A. palmata* larvae respond to such elevated temperatures. If they are negatively affected, larval supply, dispersal potential, and recruitment of this important Caribbean species could all be reduced. Here, we report the results of laboratory experiments comparing the embryogenesis and larval development, larval survival, settlement, and short-term post-settlement survival of *A. palmata* larvae cultured at temperatures from about 27.5 to 31.5 °C.

Materials and Methods

Spawn collection

Laboratory experiments were conducted in two field locations during the summers of 2007 and 2008. During both field seasons, expected dates and times of spawning were determined using the lunar cycle and previously observed spawning dates (Szmant and Miller, 2005). The first experiment took place in 2007 from 1 through 8 September at the

Instituto de Ciencias del Mar y Limnología (ICMYL) in Puerto Morelos, México. The spawn was collected from adult *Acropora palmata* colonies within the Puerto Morelos Coral Reef National Park. Two colonies provided most of the spawn used in the experiment, with a small contribution from a third colony. The second experiment was conducted in 2008 from 19 August through 3 September at the Magueyes Field Laboratory of the Department of Marine Science, University of Puerto Rico, Mayaguez (UPRM) in La Parguera, Puerto Rico. Spawn was collected from colonies in the Tres Palmas Marine Reserve in Rincón, PR. Spawn from five *A. palmata* colonies was combined with spawn collected in plankton nets from an unknown number of colonies. The density and abundance of adult *A. palmata* colonies in Rincón was higher than in Puerto Morelos. After combining the spawn from all the colonies, fertilization was allowed to take place for about an hour, after which sperm were rinsed from the eggs. The washed spawn (mostly fertilized eggs, but these cannot be distinguished or separated from unfertilized ones until cleavage begins) was transported to the laboratory by boat (México) or car (Puerto Rico). Field temperatures both years at the time of spawning were about 30 °C (not measured accurately).

Temperature selection and exposure systems

Three nominal treatment temperatures were chosen with the intent of comparing larval responses to (1) a temperature within the upper annual Caribbean temperature range that is generally considered nonstressful (28 °C); (2) a temperature that can be considered a common but stressful summer maximum (30 °C); and (3) an elevated temperature observed during major coral bleaching and mortality events over the last two decades (31.5 °C). Earlier preliminary temperature experiments with *A. palmata* embryos demonstrated high mortality at temperatures above 32 °C. Therefore, 31.5 °C was chosen as the highest level to be tested. Table 1 summarizes the temperatures actually achieved each year for each temperature treatment, which varied by 0.7 to 1.0 °C between years. For comparison, nominal target temperatures are used to describe treatments from both years from this point forward.

The temperature exposure systems consisted of twelve 12-l polycarbonate containers, each equipped with a Cool-Works Ice Probe aquarium chiller, a Pro Heat aquarium heater, and a small pump to rapidly circulate the water within the bath. A HOBO Pro V2 data logger programmed to record temperature every 10 min was placed in each bath and showed that temperatures were maintained within ± 0.2 °C (Table 1). Several times per day, temperatures were also checked with a calibrated digital thermometer and adjusted if necessary. Each temperature bath had a floating rack that suspended containers with the embryos or larvae (plastic bowls or test tubes as described below). In both years,

Table 1

Summary of experimental temperatures for both field seasons

EXPERIMENT		SURVIVORSHIP BATHS						MASS CULTURE BATHS						INCUBRIDGES					
Year	Replicate	28 A	28 B	30 A	30 B	31.5 A	31.5 B	28 A	28 B	30 A	30 B	31.5 A	31.5 B	28 A	28 B	30 A	30 B	31.5 A	31.5 B
2007	Nominal Temperature (°C)																		
	Mean	27.5	27.3	29.1	29.1	31.4	31.3	27.4	27.6	29.1	28.8	31.5	31.5						
	SD	0.11	0.09	0.18	0.04	0.06	0.20	0.10	0.43	0.13	0.20	0.06	0.06						
	Min.	27.0	27.1	24.1	29.0	31.3	31.0	27.3	26.6	28.9	27.8	31.3	31.3						
2008	Max.	27.8	27.5	29.4	29.2	31.6	32.1	27.6	28.4	29.7	29.0	31.7	31.7						
	Nominal Temperature (°C)																		
	Mean	28.2	28.1	29.8	29.9	31.8	31.7	28.2	28.4	30.0	29.9	31.7	31.5	28.2	28.0	30.0	30.0	31.5	31.4
	SD	0.07	0.22	0.17	0.11	0.04	0.06	0.06	0.10	0.04	0.12	0.03	0.27	0.07	0.17	0.20	0.21	0.28	0.23
	Min.	28.0	27.7	29.4	29.6	31.7	31.6	28.0	28.0	29.9	29.6	31.6	30.9	28.0	27.5	29.5	29.7	31.1	31.2
	Max.	28.3	28.6	30.3	30.0	31.9	31.8	28.3	28.6	30.1	30.2	31.8	32.0	28.3	28.3	30.2	30.4	32.0	31.9

There were two replicate baths per temperature for survivorship measurements, and two replicate baths per temperature for mass culture of the development samples (replicates designated A and B in each case). In 2008, IncuFridge incubators (two per temperature) were used for temperature treatments during settlement. Mean, standard deviation, minimum and maximum temperatures for each replicate temperature bath are included for each year. Standard error values were calculated for each treatment but were <0.005 for most data sets; therefore, standard deviations are presented.

temperature systems were set up inside a laboratory with standard overhead fluorescent lights without a controlled photoperiod.

Four baths were maintained at each of the three target temperatures. Two of the baths at each temperature contained three to four 500-ml plastic bowls suspended in the water bath, each loaded as soon as possible after spawning with approximately 5000 embryos or eggs (these could not be distinguished from each other until the fertilized eggs started to cleave, but fertilization success was measured to be over 90% in both years). These embryos/larvae were used to supply the survivorship tests and the development samples. The two additional baths at each temperature held 50-ml test tubes used in the survivorship tests.

Survivorship tests

Survivorship was measured by counting out 20 embryos or larvae from the culture bowls into 50-ml plastic culture tubes, and re-counting the numbers remaining in the tube after a given interval of temperature exposure. Counts were conducted twice daily during the first 2 days of exposure and daily afterward. The seawater in the tubes was replaced after each count with filtered seawater warmed to the appropriate temperature. Seawater in the culture bowls was changed twice daily. When the average number of embryos/larvae in the replicate tubes at any given temperature decreased below 10, new embryos/larvae were taken from the culture bowls at the same temperature to bring the number back to 20. Without replenishment, over time, the numbers of individuals in each sample would have dropped below 50% and any subsequent measurements of survivorship would have had high uncertainty. Especially for the highest temperatures with high mortality, few larvae would have been left for measuring survivorship after the first few counts. The new embryos/larvae added to the survivorship tubes were ones that had survived the same number of hours of temperature exposure in the culture bowls, and only healthy looking embryos/larvae at the same stage of development were chosen. While culture conditions in the bowls were initially more crowded than in the tubes, density in the bowls decreased as samples were removed for the development time series, as well as due to mortality similar to that in the test tubes. By the last 48 h of the experiment, the densities in the culture bowls at the highest temperatures were quite low (not measured, visually obvious). This approach was used instead of replacing all 20 embryos/larvae after each count because of time constraints and the limited number of embryos/larvae we could accommodate in the temperature exposure baths. Although we recognize that the replacement embryos/larvae were being cultured under initially higher densities and different conditions from those in the tubes, the fact that the best of the survivors in the bowls were being selected as replacements suggests that the sur-

vivorship rates would err on the conservative side. Survivorship was back-calculated for each tube after each replacement, and a cumulative survivorship curve was generated for each tube.

First temperature experiment, Puerto Morelos 2007

About 2 h after the end of the fertilization period, spawn was transferred into the culture bowls filled with filtered seawater to begin the temperature exposure. The seawater was from the lagoon adjacent to the ICMYL laboratory pumped into 2000-l storage containers, and then pumped through 20- μ m and 5- μ m filters before use. Field salinity was 35 and temperature was 30 °C. Initiation of the survivorship test was delayed so as to be able to distinguish developing embryos from unfertilized eggs. About 14 h after fertilization, 12 batches per temperature of 20 developing embryos in the prawn chip to early blastula stage (Fig. 1) were counted from the culture bowls and placed into the 50-ml tubes with 30 ml of 5- μ m-filtered seawater. Six tubes were then suspended in each of the two survivorship baths per temperature to begin the survivorship runs. The numbers of surviving embryos in each tube were counted twice per day for 2 days, and then once per day for the following 5 days as described above.

Second temperature experiment, La Parguera 2008

The procedures used during the 2008 temperature exposure were similar to those used in 2007. About 2 h after fertilization, spawn was placed into each of the three plastic culture bowls per temperature bath, initiating the temperature exposures. The seawater used this year was from the Magueyes running seawater system filtered through a sand filter followed by 20- μ m, 5- μ m, and 0.2- μ m filters. About 4 h after spawning, when fertilized eggs were in the 2- to 16-cell stage, 12 replicate samples of 20 developing embryos per temperature were counted from the culture bowls into the tubes with 30-ml of 0.2- μ m-filtered seawater and placed as per the previous experiment, six per temperature bath. Salinity from the running seawater system varied between 35 and 36. For the following 7 days, the numbers of survivors were counted as described above.

On the eighth day following spawning, the planulae surviving in the 50-ml tubes at each temperature were combined, and 12 samples per temperature of 10 larvae were counted into the wells of 6-well culture plates with 10 ml of 0.2- μ m-filtered seawater. A small chip, ~ 0.5 cm², of an unidentified species of crustose coralline algae (CCA), was placed in half of the wells at each temperature. All CCA chips were generated from one individual piece of alga. The remaining wells had no added substrate and served as settlement controls. The culture plates were kept at the experimental temperatures in two small incubators per temperature (Incufridge model RS-IF-202) for settlement. Each

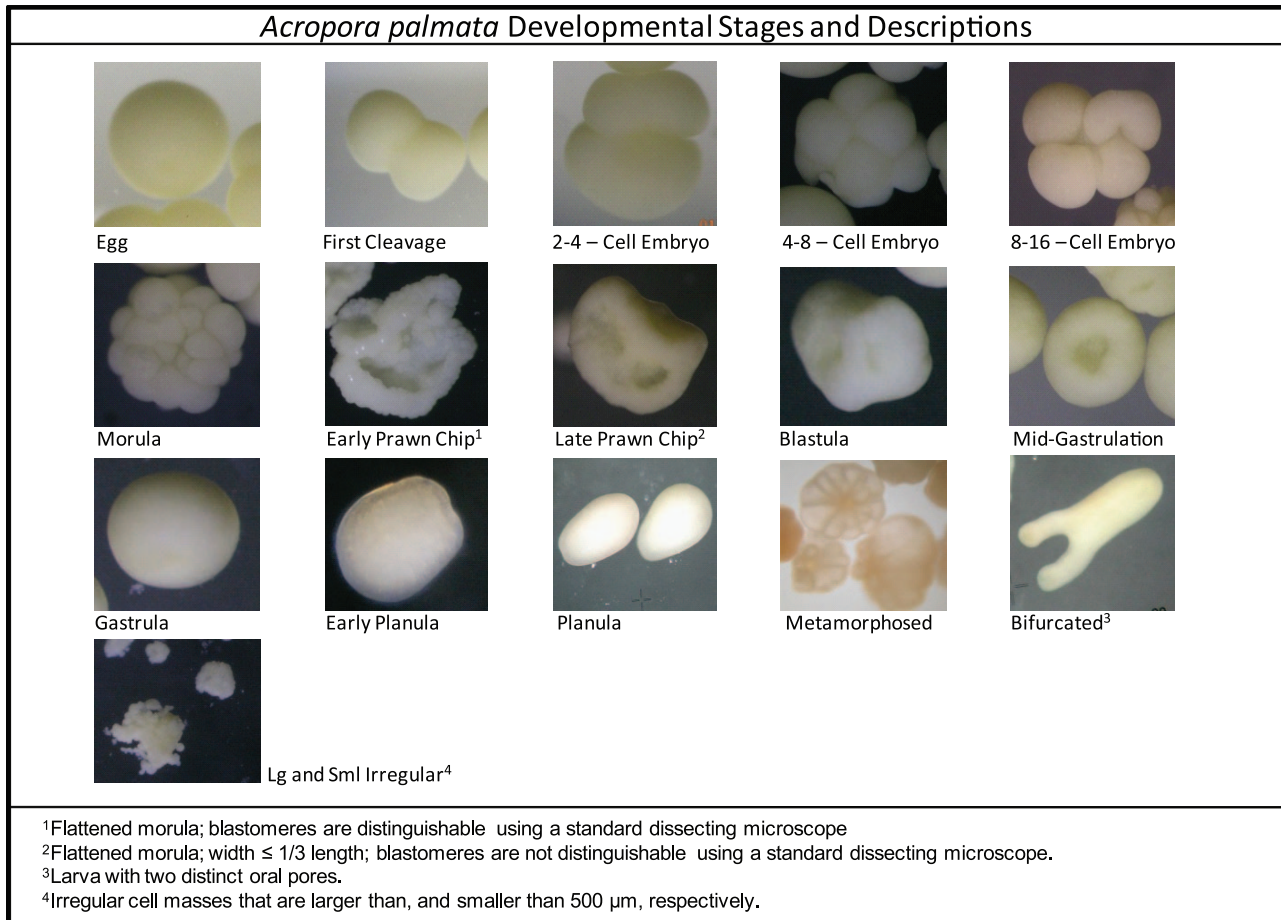


Figure 1. Photomicrographs of glutaraldehyde-fixed or live embryos and larvae arranged to show the developmental stages of *Acropora palmata*. Embryos and larvae ranged from about 550 to 900 μm in diameter (average diameter 700 μm).

Incufridge was equipped with a fan to help even the temperature within the chamber. The samples were examined about 22, 44, and 60 h after the CCA chip was added and the numbers of larvae that had settled were counted. The water in each well was replaced with preheated water every 24 h.

After the 60-h settlement experiment, all non-settled larvae were removed from the wells. CCA chips with settlers were returned to their wells with clean, filtered water at temperature, and survivorship was assessed daily for the following 5 days.

Developmental time series

Both summers, during the first 2 days after fertilization, samples of about 100 embryos were taken about every 4 h from the mass culture containers and fixed in 2% glutaraldehyde in 0.05 mol l⁻¹ sodium cacodylate buffer. For the next 6–7 days, samples were collected every 6–12 h. Preserved samples were returned to UNCW and examined using a dissecting microscope at magnifications up to 90 $\times$ to identify and quantify developmental stages. The 17 developmental stages

used here are based on the 14 stages described by Okubo and Motokawa (2007) for five Pacific species of *Acropora*, plus three additional stages we defined for our experiment: bifurcated larvae (those with two oral pores), irregularly developing larvae, and prematurely metamorphosed polyps (Fig. 1). The percentage of embryos/larvae in each developmental stage in each sample was calculated from the total number of all normally developing embryos/larvae counted in that sample. The number of abnormal embryos/larvae in each sample was also quantified. The number of embryos/larvae in each sample averaged 82 larvae (± 6 s.e.; min = 10; max = 194) in 2007 and 89 (± 6 s.e.; min = 21; max = 257) in 2008. Late in each exposure period at the highest temperatures, sample sizes were smaller because the numbers of larvae available for sampling were low due to high mortality.

Data analysis

Cumulative survivorship was calculated for each replicate tube at each time point, taking into account when

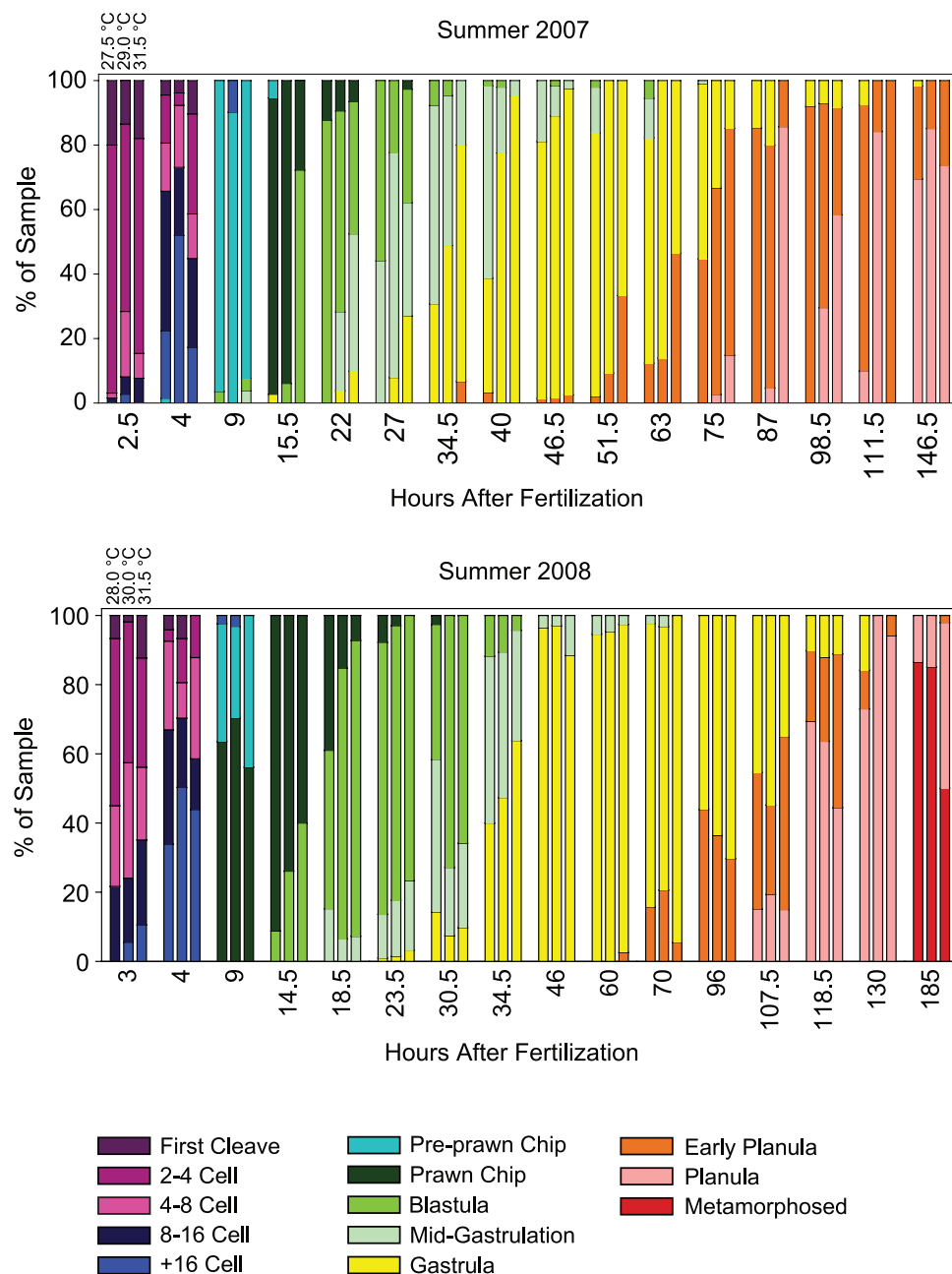


Figure 2. Comparison of the time course of development of *Acropora palmata* embryos and larvae at three nominal temperature treatments, 28 °C (control), 30 °C (intermediate), and 31.5 °C (elevated). Each group of three bars represents the percentage of the various stages present in samples from larval cultures at each of the three temperatures at any given sampling time. Developmental time series from the 2007 experiment (upper panel); the 2008 experiment (lower panel). Note: The x-axes represent categorical time points and are not linear. Percentages were calculated using only those embryos exhibiting normal stages of development.

samples were replenished. Percent data were transformed using the arcsin square root or a modified logit equation to produce data sets with acceptable normality and equality of variance. The replenishment procedure precluded the use of a repeated measures analysis of variance (ANOVA), and thus a single-factor ANOVA was used to compare survivorship among temperatures at individual time points. A

Bonferroni correction to the *P* values was applied to correct for multiple comparisons. Two-factor repeated-measure ANOVAs were used to compare larval settlement over time among temperatures. Single-factor ANOVAs were used to examine differences in post-settlement survivorship. SigmaStat 3.5 and SAS 9.1.3 were used for ANOVA computations.

Table 2

Statistical comparisons of developmental stage distributions of *Acropora palmata* embryos and larvae cultured at three temperatures during two field experiments

Hours after fertilization	2007 DEVELOPMENT				2008 DEVELOPMENT				
	29 °C compared to 27.5 °C		31.5 °C compared to 27.5 °C		Hours after fertilization	30 °C compared to 28 °C		31.5 °C compared to 28 °C	
	Chi-squared	P value	Chi-squared	P value		Chi-squared	P value	Chi-squared	P value
2.5	128.0	0.003*	15.3	0.105	3.0	25.3	0.010*	82.3	0.002*
4.0	33.7	0.006*	9.0	0.841	4.0	77.4	0.002*	16.2	0.044*
9.0	57.3	0.006*	7.0	0.544	14.5	16.0	0.005*	39.8	0.002*
15.5	11.3	0.688	1406.5	0.001*	18.5	18.9	0.005*	43.8	0.002*
22.0	844.2	0.003*	1535.1	0.001*	30.5	38.6	0.002*	11.8	0.141
27.0	89.6	0.003*	747.7	0.001*	34.5	0.8	1.000	22.6	0.002*
40.0	37.4	0.003*	62.7	0.001*	185.0	0.6	1.000	60.3	0.004*
75.0	24.9	0.012*	§	§					
87.0	25.8	0.032*	§	§					
98.5	939.3	0.003*	§	§					
146.5	57.7	0.003*	§	§					

Chi squared analyses with a Monte Carlo technique were used. Distributions of larvae at the nominal temperature of 28 °C were designated as the expected distributions, and the distributions at the nominal temperatures of 30 and 31.5 °C were compared to the expected. Bonferroni corrected *P* values are presented.

* Effect significant at 0.05 level. Only time points at which there were significant differences are presented.

§ Time points at which no statistical analyses were run due to samples with fewer than 50 individuals.

The chi-squared statistic was used to analyze differences in developmental stage distributions at each time point. The distribution of embryos/larvae among stages at 28 °C was designated as the expected distribution, and the distributions at the two higher temperatures were compared against it. Only samples containing 50 or more embryos and larvae were used in the analysis. Because there was only a single sample per temperature per time point, *P* values were simulated using a Monte Carlo technique, and Bonferroni *P* value corrections were applied to correct for multiple comparisons. A continuity correction was also applied to account for occasional expected values of 0 at 28 °C. The statistics program R, ver. 2.8, was used for chi-squared analyses. Differences were considered significant at *P* < 0.05.

Results

Embryogenesis and larval development

The time-course of development for *Acropora palmata* at 28 °C was similar to that described by Okubo and Motokawa (2007) for their Pacific species of *Acropora* maintained at 26–26.5 °C. *A. palmata* embryos at the two higher temperatures moved through the developmental sequence faster than those cultured at 28 °C (Fig. 2; Table 2). In the 2007 temperature experiment, the pattern of accelerated development was significant at the earliest sample time at

30 °C, but not at 31.5 °C. By 15 h after fertilization 72% of the embryos at 31.5 °C had reached the blastula stage, compared to only 6% at the 30 °C and 0% at 28 °C (*P* = 0.001; Fig. 2, top panel). This pattern of accelerated development continued 22 h after fertilization when 42% of the blastulae at 31.5 °C had started to invaginate; this process was not observed at 28 °C until 27 h after fertilization (*P* = 0.001). Gastrulae, early planulae, and fully developed planulae all appeared earlier in the time course at 30 and 31.5 °C than at 28 °C. The acceleration was most obvious 87 h after fertilization when 86% of the larvae had reached the planula stage at 31.5 °C and none had reached that stage at 28 °C.

In the 2008 temperature experiment, the pattern of accelerated development at the higher temperatures was less pronounced but still present (Fig. 2, lower panel). Fifteen hours after fertilization, 40% of the embryos at 31.5 °C were in the blastula stage compared to only 9% at 28 °C (*P* = 0.002). Gastrulae and early planulae stages were also observed sooner at 30 and 31.5 °C than at 28 °C, but statistical analyses could not be run due to small sample sizes. During this year, fully developed planulae were observed at the same time at the three temperatures.

In addition to an acceleration in development, a greater percentage of embryos and larvae exhibited abnormal development at the elevated temperatures (Fig. 3). During the period between 3 h and 52 h in the 2007 experiment, the average number of irregularly developing and misshapen embryos was twice as high at 31.5 °C as at 28

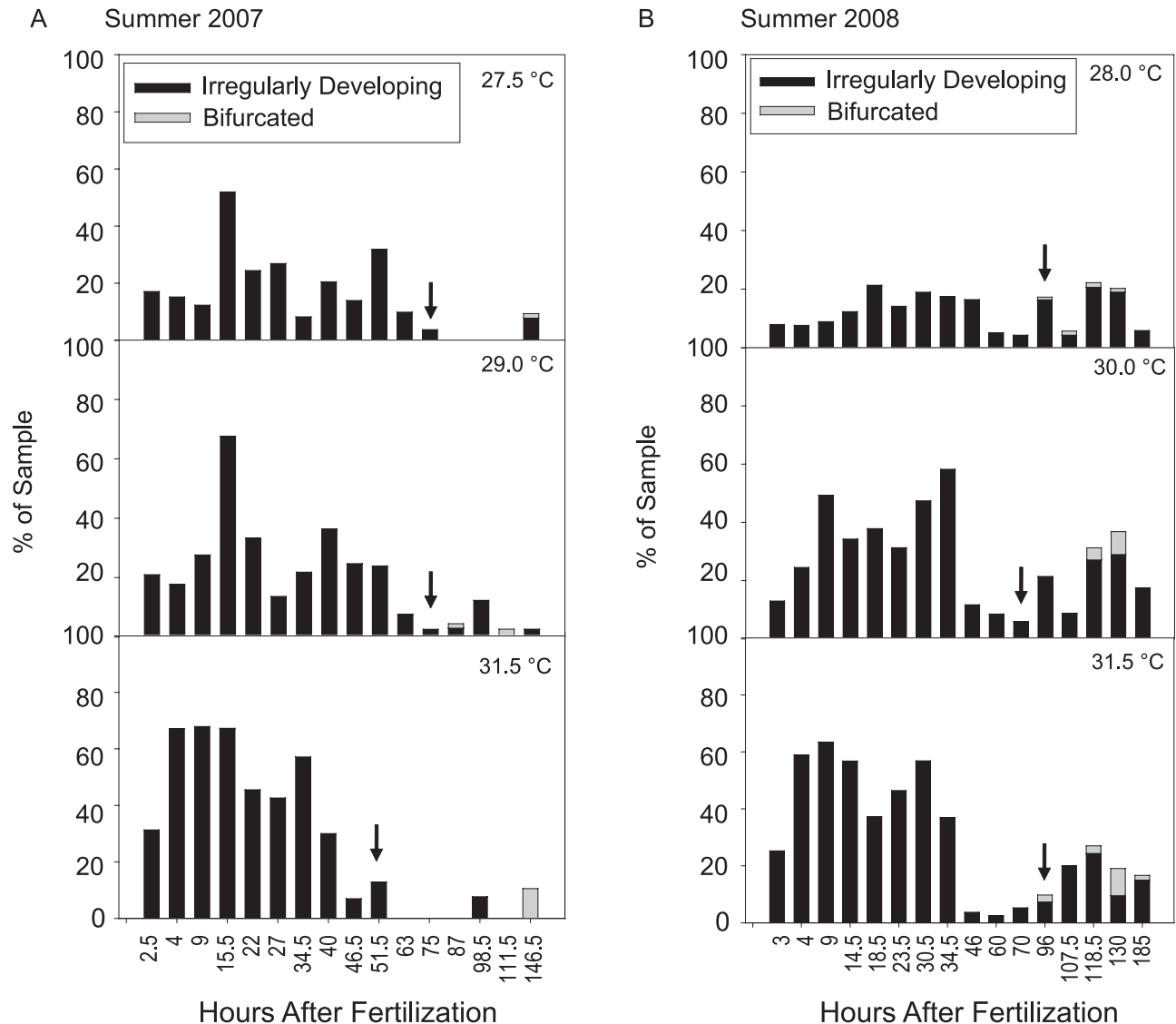


Figure 3. Percentage of irregular *Acropora palmata* embryos/larvae in each developmental sample at each of three nominal temperature treatments. Percent of abnormally developing embryos (A) during 2007; (B) during 2008. Arrows indicate time points when >20% of the sample at a given temperature was in the early planula stage. The x-axes represent categorical time points and are not linear.

°C (43% compared to 22%). In the 2008 experiment, over that same time frame, 37% of the samples at 31.5 °C were developing abnormally compared to 11% at 28 °C, a 3-fold increase in developmental aberrations. In both years, the percentage of abnormally developing embryos decreased after gastrulation, once the larvae reached the planula stage. The planula stage was reached sooner at the elevated temperatures in 2007 than in 2008 (Fig. 3: first appearance of planulae at more than 20% of sample indicated by arrows). During the 2008 experiment, the percentage of developmental abnormalities at the higher temperatures increased again toward the end of the exposure period.

Survivorship

Embryos and larvae at the two higher temperatures exhibited a significantly reduced survivorship compared to that at 28 °C, with the lowest survivorship at 31.5 °C both years ($P < 0.000$; Fig. 4A). In 2007, only 4% of the larvae at 31.5 °C survived to the end of the 140-h temperature exposure, a 5-fold decrease from an average survivorship of 19% at 30 °C, and an 8-fold decrease from the average survivorship of 34% at 28 °C ($P < 0.000$; Fig. 4A). This same pattern was observed in 2008, but the overall survivorship in each treatment was higher. Survivorship after the 170-h temperature exposure at 31.5 °C was 9%, whereas at

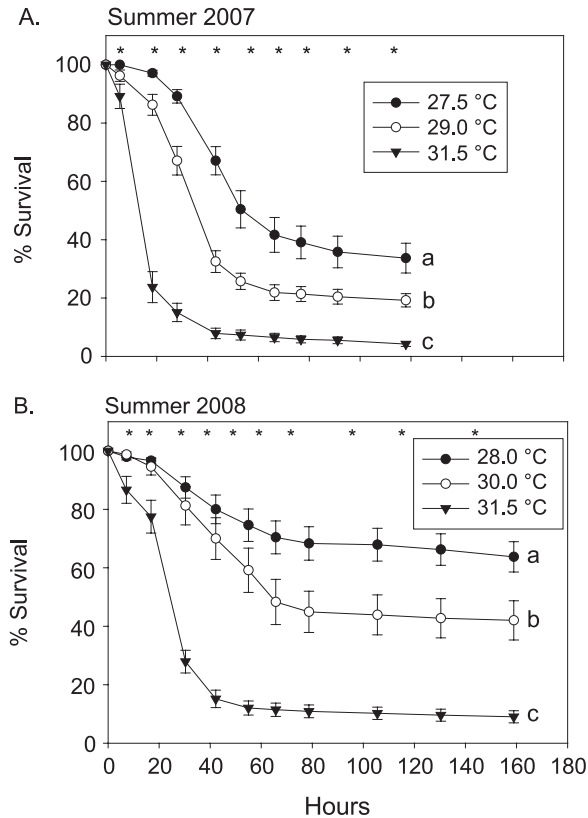


Figure 4. Percent survivorship of *Acropora palmata* embryos or larvae exposed to one of three nominal temperatures during development. (A) Survivorship curves from 2007. Time 0 indicates the start of the survivorship experiment, which was 14 h after the start of temperature exposure and 16 h after fertilization. (B) Survivorship curves from 2008. Time 0 indicates the start of the survivorship experiment, which was 2 h after the start of temperature exposure and 4 h after fertilization. $n = 12$; error bars = s.e. Letters a, b, and c indicate statistically significant differences among temperatures at $P < 0.05$, and asterisks indicate each time point in which differences among temperatures were significant at $P < 0.05$.

28 °C and 30 °C, survivorship averaged 64% and 42%, respectively ($P < 0.000$; Fig. 4B). There was a clear pattern of survivorship decreasing with increasing culture temperature (Fig. 5).

Embryos cultured at all three temperatures, both years, experienced a decrease in survivorship between 15 and 80 h after fertilization (Fig. 6). In 2007, this decrease occurred about 60 h after fertilization at 28 °C, whereas at 31.5 °C it occurred earlier, between 25 and 50 h (Fig. 6A; compare top and bottom panels). At 30 °C, the decreased survivorship occurred at about 50 h. Furthermore, the magnitude of the survivorship decrease was greater as the culture temperature increased (Fig. 6A). After these decreases, survivorship increased and leveled off near 100% for the remainder of the experiment (Fig. 6A). A similar trend was observed during the second year (Fig. 6B).

The embryonic stages at which these decreases in survi-

vorship occurred corresponded to stages associated with gastrulation, as evidenced by invaginating embryos and gastrulae (Fig. 6). Since survivorship increased as gastrulae developed into planulae, the leveling off of survivorship occurred earlier at the higher temperatures. Once the planula stage was reached, mortality decreased, on average to less than 10% per day (Fig. 6).

Settlement and post-settlement survivorship

Settlement decreased with increasing temperature ($P = 0.015$; Fig. 7). At 28 °C, mean settlement was 62%, compared to 43% at 30 °C and 37% at 31.5 °C. Almost all of the larvae that settled did so within the first 22 h after induction. No metamorphosis or settlement was observed in the settlement wells without CCA. No differences in post-settlement survivorship were observed among temperatures over the following 4 days, and post-settlement survival was high for all treatments (88%–100%).

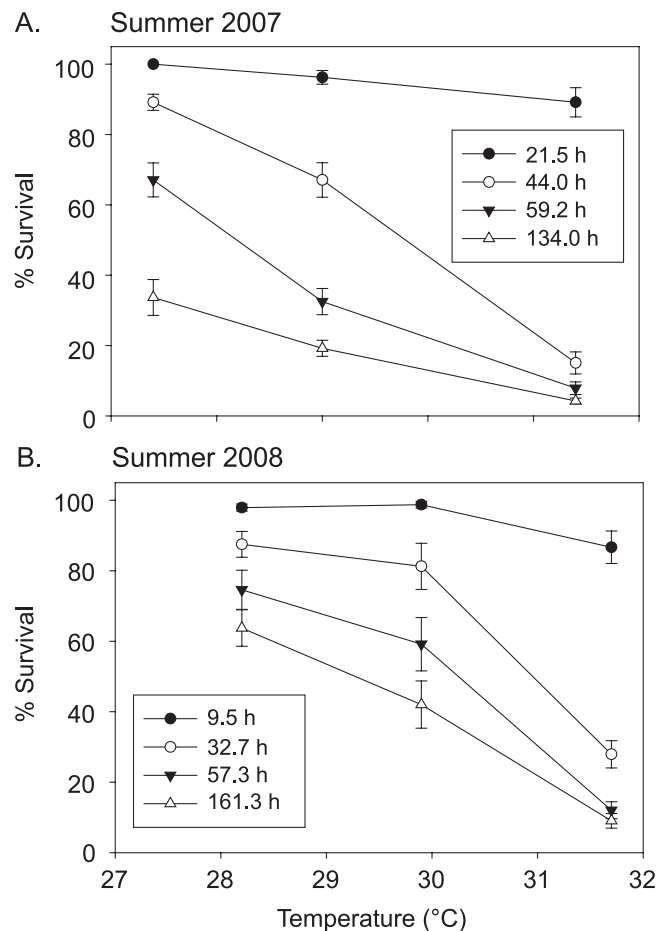


Figure 5. Percent survivorship of *Acropora palmata* embryos or larvae at select times during the survivorship experiments, plotted against exposure temperature. (A) 2007 experiment; (B) 2008 experiment. $n = 12$; error bars = s.e.

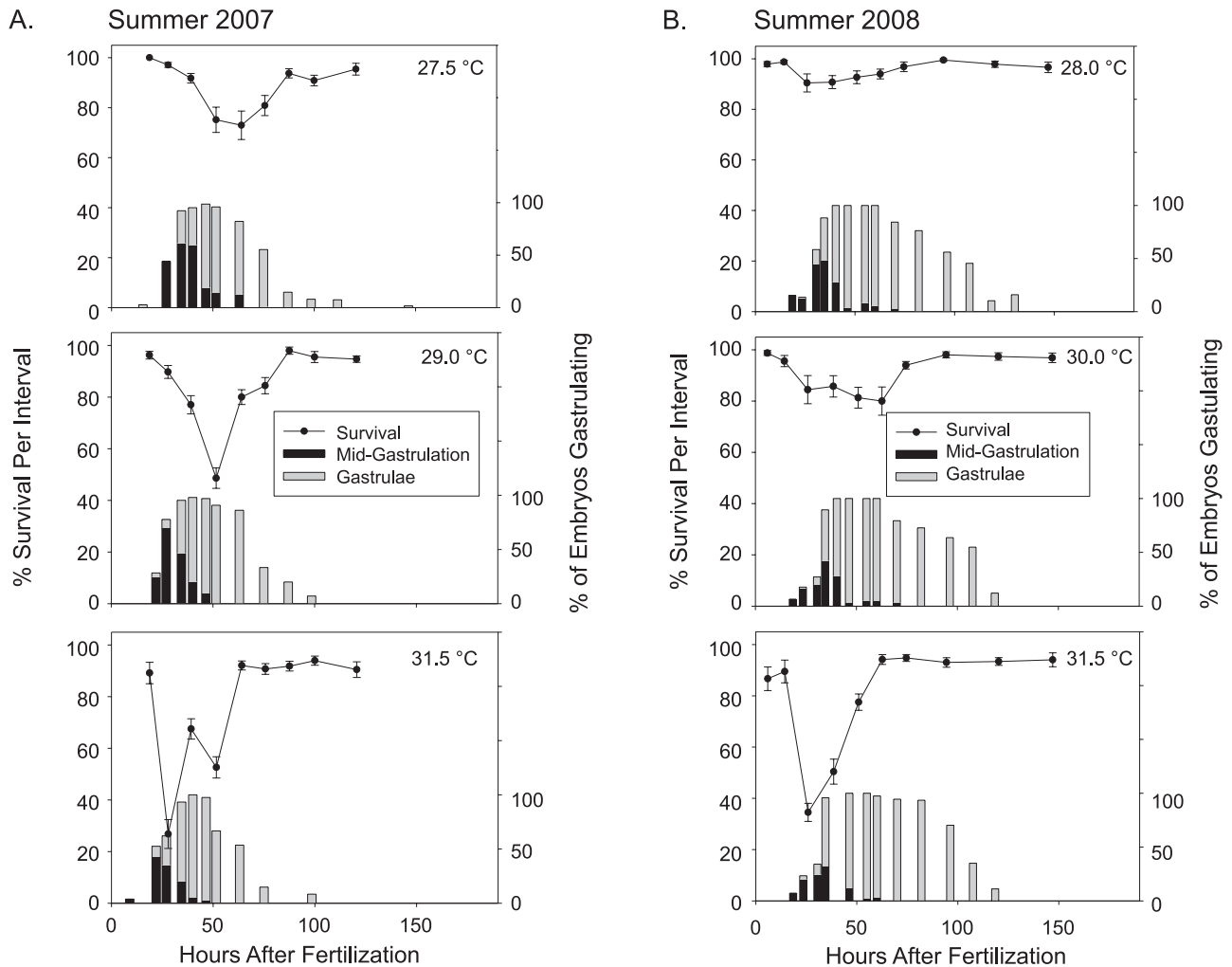


Figure 6. Percent survival per sampling interval of *Acropora palmata* embryos and larvae exposed to each of three nominal temperatures during development. The percentages of the embryos in gastrulating developmental stages are also shown. The percent survival data for each sampling interval is plotted against the mid-time of that interval. (A) 2007 experiment; (B) 2008 experiment. Error bars = s.e.

Discussion

This study tested the response of *Acropora palmata* embryos and larvae to temperature conditions that have been experienced in the Caribbean Sea during recent record warm summers (e.g., 1998, 2005; Winter *et al.*, 1998; A. Szmant, pers. obs.) and which are expected to occur more frequently as the climate warms over the next century. Our results demonstrate a potentially severe reduction in the reproductive success of the important reef-builder and threatened species *A. palmata* as a result of negative temperature effects on embryonic development, survivorship, and larval settlement. Most significant was the reduction in survivorship throughout the development period, resulting in as much as a 5-fold decrease in survivorship caused by a temperature increase of only 1.5 °C over an average sum-

mer high of 30 °C. This increased mortality could severely reduce the supply of *A. palmata* larvae to Caribbean reefs.

Previous studies have shown that elevated temperatures can not only cause high mortality during development, but can also negatively affect gametogenesis, fecundity, and fertilization, and may cause eggs released under temperature stress to lyse (Jokiel and Guinther, 1978; Szmant and Gassman, 1990; Bassim, 1998; Omori *et al.*, 2001; Bassim *et al.*, 2002; Krupp *et al.*, 2006). In addition, elevated temperatures can cause developmental aberrations and reduce settlement success and/or post-settlement survival (Jokiel and Coles, 1990; Edmunds *et al.*, 2001; Bassim *et al.*, 2002; Bassim and Sammarco, 2003; Krupp *et al.*, 2006; Randall and Szmant, 2009; this study). Negative effects of elevated temperatures acting on every stage of reproduction and embryonic/larval development could result in a critical fail-

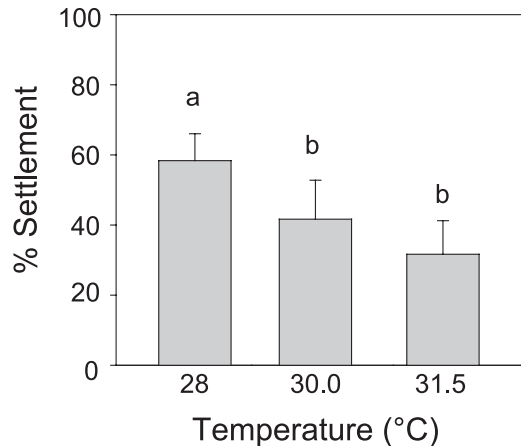


Figure 7. Percent settlement of *Acropora palmata* larvae, 22 h after settlement induction, at each of three nominal treatment temperatures. $n = 6$; error bars = s.e. Letters a and b indicate statistically significant differences among temperatures; differences were significant at $P < 0.05$.

ure of supply-side-related recruitment for this species. For example, if a temperature-stressed adult *A. palmata* population exposed to 31.5 °C experienced a 50% reduction in reproductive output, followed by 75% mortality during larval development, and a 50% reduction in settlement, the resulting recruitment would be only 6% of the “normal” potential. This 6% does not take into account the reduction in the number of adult colonies that has already occurred, due to disease, hurricane damage, and other factors, which further reduces reproductive potential.

Although our results from 2007 and 2008 exhibited the same response pattern to elevated temperature, survivorship at all temperatures during 2008 was higher than in 2007: at the lowest temperatures (28.0 °C) in 2008 it was 64% compared to 34% in 2007. The observed lower survivorship in the 2007 experiment could be due to low genetic variation, gamete viability, poor health of the Mexican corals, differences in other stressors influencing the populations, or unintentional differences in experimental conditions outside of temperature control. During the 2007 experiment, the majority of the spawn came from only two colonies from a low-density population, most of which did not spawn. The spawn for the 2008 experiment came from 5+ colonies from a large, genetically diverse population (Baums *et al.*, 2005) that underwent massive spawning, and thus the test larvae came from a much more diverse genetic pool. The higher survivorship observed in 2008 may indicate that some larvae are less sensitive to elevated temperatures than others and there may be the potential for adaptation to elevated temperatures over time.

Our results also demonstrate that elevated temperatures negatively affect the physiology of the animal host directly, rather than merely causing indirect negative effects mediated by temperature effects on endosymbiotic algae, which are absent in *A. palmata* larvae. These results reinforce

findings of Baird *et al.* (2006), who demonstrated that both symbiotic and nonsymbiotic larvae of *A. muricata* responded similarly to temperature exposure regimes. This is also supported by Bassim *et al.* (2002) and Bassim and Sammarco (2003), who demonstrated negative effects of elevated temperatures on larvae of *Diploria strigosa*, which are also azooxanthellate. While much thought has been given to the possibility that adult corals may acclimatize and adapt to increasing temperatures *via* changes in their symbiotic *Symbiodinium* populations (Glynn, 2004), this is not a possible mechanism in asymbiotic coral larvae. Furthermore, the acclimatization and adaptation of adults to elevated temperatures will have little evolutionary benefit if their reproductive stages cannot survive to contribute to the next generation.

A. palmata embryos exposed to elevated temperature exhibited obvious developmental irregularities including misshapen and disorganized cells, indicating that on a whole-embryo scale, cell re-arrangement and migration was likely not occurring correctly. However, the effects of elevated temperature at the ultrastructural level are unknown. Our results indicate that gastrulation was the most temperature-sensitive stage of *A. palmata* embryogenesis. The physiological responses of developing corals to elevated temperatures are not well known, but the high sensitivity of gastrulation to temperature has been documented in other species and is, therefore, not surprising to find in corals (Cossins and Bowler 1987). *A. palmata* embryos appeared to gastrulate *via* invagination, as has been described for other *Acropora* spp. (Ball *et al.*, 2002; Okubo and Motokawa, 2007; Hirose *et al.*, 2008). Invagination includes many processes that elevated temperature can disrupt, such as gene expression, cell re-arrangement, and cell differentiation (Cossins and Bowler, 1987; Byrum and Martindale, 2008). Other possible general effects of elevated temperature during development include inhibited signaling pathways, arrested mitotic divisions, impairment of functional enzymes and structural proteins, and disruption of cell migration and communication (Negri *et al.*, 2007).

One possibility is that elevated temperature negatively affects transcription and/or translation of embryonic genes just prior to and during gastrulation. Because early embryogenesis is controlled by maternal morphogenic factors that are provided to the embryo *via* the egg (Gilbert, 1991; Byrum and Martindale, 2008), transcription of embryonic genes is generally not required during early cleavage, morula and blastula development. However, as embryos mature, transcription of embryonic genes is required to regulate later embryogenesis. Elevated temperatures could affect regulation and transcription of genes involved in transcription itself, as well as those important to cell replication, cytoskeletal/cell-adhesion processes, stress responses, and protein synthesis. A number of pathways are involved in regulation of development, and recently *Wnt*

signaling molecules, a highly conserved family of genes that control gastrulation and germ layer specification during embryonic development, have been identified in cnidarians (Kusserow *et al.*, 2005; Guder *et al.*, 2006; Lee *et al.*, 2006; Byrum and Martindale, 2008; Momose *et al.*, 2008). Disruption by elevated temperatures of this pathway, as well as of a number of other pathways important to gastrulation, could have contributed to the high mortality of *A. palmata* embryos during gastrulation. A microarray approach such as that reported by Grasso *et al.* (2008) should help elucidate which gene systems are affected by elevated temperatures.

Despite the lack of understanding of the mechanisms by which elevated temperature caused increased developmental abnormalities and mortality, reproductive output and recruitment potential of the important Caribbean coral *A. palmata* are likely being severely reduced at the levels of elevated temperatures experienced by populations in the northern Caribbean in recent years. With the projected estimates of an increase in global temperatures of 1.8 to 4 °C by 2100 and a corresponding predicted increase in tropical sea surface temperatures (Bates *et al.*, 2008), recovery of *A. palmata* populations by sexual recruitment will be unlikely.

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